

Notes: Gupta and Hansman (RFS, 2022)

Selection, Leverage, and Default in the Mortgage Market

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1 Big picture

- **Question.** Why are highly leveraged mortgage borrowers more likely to default? Is the leverage-default correlation mostly a **causal effect of leverage itself** (moral hazard), or does it mainly reflect **ex ante risky borrowers choosing larger loans** (adverse selection)?
- **Setting.** The paper studies **option adjustable-rate mortgages (option ARMs)** originated in the U.S. between 2004 and 2007. These contracts had fixed minimum payments in the early years but floating rates tied to mortgage indexes such as LIBOR or Treasury.
- **Core challenge.** In standard mortgage data, initial leverage and realized leverage move together, so it is hard to tell whether default is driven by the borrower's original contract choice or by the realized home equity position later on.
- **Main empirical idea.** The unexpected **2008 divergence between LIBOR and Treasury rates** created ex post variation in borrowers' balances that was not salient when they chose their loans. Because option ARM payments were fixed in the short run, index-driven rate changes moved **balances** much more than they moved **current required payments**.
- **Identification logic.**
 - **Moral hazard** is identified from borrowers who made the *same initial leverage choice* but later faced different realized leverage because their mortgage indexes evolved differently.
 - **Adverse selection** is identified from borrowers who made *different initial leverage choices* but are compared at similar realized leverage once the exogenous balance variation is netted out.
- **Main reduced-form specification.** The paper regresses default between months 24 and 36 on both **origination leverage** and **current leverage**, instrumenting current leverage with a

leave-one-out mean of current LTV for borrowers with the same **index type** $\times$ **origination month**.

- **Main result.** In the baseline decomposition, **adverse selection explains about 60%** of the leverage-default correlation in the sample, while **moral hazard explains about 40%**.
- **Key magnitude.** A **10-point reduction in current LTV** at month 24 lowers the probability of default between months 24 and 36 by about **4.7 percentage points** for compliers in the IV specification.
- **Key nonlinearity.** The causal effect of leverage is strongest **around the negative-equity threshold**. Default risk jumps sharply once borrowers move into negative equity, but becomes much flatter for borrowers already far underwater.
- **Joint-model result.** A control-function model also finds significant adverse selection, with an estimated correlation of about $\rho_{\varepsilon v} = 0.12$ between unobserved default risk and leverage demand.
- **Policy result.** A regulator who ignores adverse selection will **overstate the default reduction** from loan-to-value caps and **understate the equilibrium costs** from higher interest rates and smaller loans.
- **Economic takeaway.** In this market, leverage is not just a source of ex post default incentives. It also acts as a **screening device**. That means leverage regulation changes both borrowers' default incentives and the menu of contracts lenders can profitably offer.

2 Data and measurement (key objects)

2.1 Why option ARMs are useful for identification

Option ARMs have four product features that matter for the paper's design:

- **Fixed minimum payment schedule.** The minimum payment is fixed initially and usually rises only by a preset amount once per year.
- **Monthly interest-rate resets.** The mortgage rate updates much more frequently than in standard hybrid ARMs.
- **Negative amortization.** If accrued interest exceeds the minimum payment, the unpaid interest is added to the balance.
- **Menu of payment options.** Borrowers are shown higher suggested payments, but in practice most pay the minimum.

Interpretation. The key feature is that **interest-rate changes shift the loan balance without immediately shifting the required payment much**. That is what lets the paper move current leverage without mechanically moving short-run payment burden.

2.2 Data source and sample construction

The data come from a loan-level panel of privately securitized mortgages provided by **Moody's Analytics / Blackbox Logic**, covering more than 90% of nonagency residential mortgage-backed

securities. The dataset contains:

- origination characteristics of the borrower, property, and contract;
- monthly information on balances, payments, delinquency, foreclosure, and modifications.

The primary sample consists of roughly **600,000 option ARMs** originated between **2004 and 2007** with origination combined loan-to-value ratios between **50 and 100**. The main estimation sample for the core IV results is the cross-section of loans **active at 24 months**, which contains **491,215 observations**.

Interpretation. The paper is not using the full U.S. mortgage market. It is using a large and important segment of **nontraditional mortgages** that was central during the housing boom.

2.3 Index types and descriptive facts

The main mortgage indexes in the sample are:

- **Treasury-indexed loans:** just over **80%** of the sample;
- **LIBOR-indexed loans:** just over **15%**;
- **COFI and fixed-rate initial periods:** each less than **2%**.

Summary statistics are broadly balanced across indexes:

- origination combined LTV averages about **0.80** for Treasury loans and **0.82** for LIBOR loans;
- average FICO scores are about **706** for Treasury and **717** for LIBOR;
- no/low documentation is extremely common, around **82–85%**.

Default rates are high in this market: about **50%** of borrowers become seriously delinquent at some point within the first five years.

Interpretation. Option ARM borrowers are not a classic low-credit-score subprime sample. They often look relatively strong on observables, but they are using a risky, highly leveraged product with weak documentation.

2.4 Key variables

The paper distinguishes carefully between **initial leverage** and **realized leverage**.

- **Origination leverage:**

$$L_i = \text{origination combined LTV.} \tag{1}$$

This uses combined loan-to-value so that junior liens are included at origination.

- **Current leverage:**

$$LTV_{it} = \text{current LTV on the option ARM at loan age } t. \tag{2}$$

The current home value is imputed using Zillow zip-code-level home price indexes.

- **Default outcome:**

$$D_{i,t+1} = 1 \quad \text{if borrower } i \text{ becomes at least 60 days past due between } t \text{ and } t + 1. \quad (3)$$

The main specification studies default between **24 and 36 months**. The paper also checks 90-day-past-due outcomes and longer horizons.

Interpretation. The entire decomposition rests on separating **what the borrower chose at origination** from **what leverage the borrower actually faces later**.

2.5 The instrument

For borrower i with index type $I(i)$ and origination month $m(i)$, the paper defines the leave-one-out mean instrument:

$$Z_{it} = \frac{1}{n_{I(i) \times m(i)} - 1} \sum_{j \neq i} LTV_{jt}, \quad (4)$$

where the sum is over all other borrowers with the same index type and origination month.

The main idea is that the **interaction of index and origination month** determines a borrower's path of realized interest rates, and hence the path of the balance.

Interpretation. Z_{it} is a compact way to isolate the exact source of exogenous balance variation the paper wants: **index-type $\times$ cohort-specific shocks to balances**.

2.6 What makes this variation plausible

Two institutional facts support the design:

- borrowers were often poorly informed about their exact mortgage index;
- much of the variation in index choice came from **lender specialization** rather than from salient borrower demand.

But the paper does *not* need random assignment to index type. It relies instead on the claim that, after controlling for index fixed effects, origination month fixed effects, and even **originator $\times$ origination month fixed effects**, the residual interaction-driven variation in current leverage is plausibly exogenous.

3 Models and empirical specifications (all equations + interpretation)

3.1 Definitions of moral hazard and adverse selection

The paper adopts reduced-form definitions tailored to mortgage default:

- **Moral hazard:** holding everything else fixed, a higher realized mortgage liability causes more default.

- **Adverse selection:** borrowers who are unobservably riskier ex ante choose higher leverage contracts.

Interpretation. These definitions map directly into different empirical predictions. Moral hazard loads on **current leverage**; adverse selection loads on **the initial leverage choice**.

3.2 Default threshold model

At loan age t , borrower i defaults if the mortgage balance exceeds the home value by more than the borrower's private cost of default:

$$B_{it} - H_{it} > C_{it}. \quad (5)$$

Dividing by home value gives a threshold rule in LTV space. After separating observables and unobservables, the paper writes the default equation as

$$D_{i,t+1} = 1\{\alpha LTV_{it} + x'_i\beta + \varepsilon_{it} > 0\}. \quad (6)$$

Interpretation. α is the moral-hazard parameter. If $\alpha > 0$, then larger current leverage causally raises the probability of default.

3.3 Leverage demand at origination

Rather than fully modeling mortgage choice in the empirical section, the paper writes a simple leverage-demand equation:

$$L_i = x'_i\psi + v_i. \quad (7)$$

Here v_i is the unobserved part of leverage demand.

Interpretation. Adverse selection is present if borrowers with high v_i are also more likely to default for unobservable reasons, that is, if

$$\rho = \text{Corr}(v_i, \varepsilon_{it}) > 0. \quad (8)$$

3.4 Reduced-form decomposition: default on current and initial leverage

The core estimating equation is a linear probability model:

$$D_{it+1} = \alpha LTV_{it} + \gamma L_i + x'_i\beta + \zeta_{j(i)} + \lambda_{I(i)} + \theta_{o(i)} \times \mu_{m(i)} + e_{it}. \quad (9)$$

The first stage is

$$LTV_{it} = \delta Z_{it} + \eta L_i + x'_i\pi + \zeta_{j(i)} + \lambda_{I(i)} + \theta_{o(i)} \times \mu_{m(i)} + u_{it}. \quad (10)$$

where:

- $\zeta_{j(i)}$ are zip-code fixed effects;
- $\lambda_{I(i)}$ are index-type fixed effects;

- $\theta_{o(i)} \times \mu_{m(i)}$ are originator $\times$ origination-month fixed effects in the most saturated specification.

Interpretation.

- α identifies **moral hazard**: the causal effect of realized leverage on default.
- γ identifies **adverse selection**: holding realized leverage fixed, borrowers who chose larger loans initially are still more likely to default.

3.5 Nonlinear specification

Because the effect of leverage on default may not be linear, the paper also estimates a version that replaces LTV_{it} with quartile dummies for current leverage and replaces L_i with quartile dummies for origination leverage:

$$D_{it+1} = \alpha_1 \mathbf{1}\{79 < LTV_{it} < 95\} + \alpha_2 \mathbf{1}\{95 < LTV_{it} < 115\} + \alpha_3 \mathbf{1}\{115 < LTV_{it}\} \quad (11)$$

$$+ \gamma_1 \mathbf{1}\{75 < L_i < 81\} + \gamma_2 \mathbf{1}\{81 < L_i < 90\} + \gamma_3 \mathbf{1}\{90 < L_i\} \quad (12)$$

$$+ x'_i \beta + \zeta_{j(i)} + \lambda_{I(i)} + \theta_{o(i)} \times \mu_{m(i)} + e_{it}. \quad (13)$$

The corresponding quartiles of Z_{it} instrument for the quartiles of current leverage.

Interpretation. This specification asks whether the causal effect is concentrated near the **negative-equity threshold** rather than spread smoothly over the whole LTV distribution.

3.6 Joint control-function model

To go beyond the linear decomposition, the paper estimates the three-equation system jointly:

$$D_{it+1} = 1\{\alpha LTV_{it} + x'_i \beta + \varepsilon_{it} > 0\}, \quad (14)$$

$$L_i = x'_i \psi + v_i, \quad (15)$$

$$LTV_{it} = \delta Z_{it} + \eta L_i + x'_i \pi + u_{it}. \quad (16)$$

The error structure is assumed to be jointly normal,

$$\begin{pmatrix} \varepsilon_{it} \\ v_i \\ u_{it} \end{pmatrix} \sim N \left(\begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 & \rho_{\varepsilon v} \sigma_v & \rho_{\varepsilon u} \sigma_u \\ \rho_{\varepsilon v} \sigma_v & \sigma_v^2 & \rho_{vu} \sigma_v \sigma_u \\ \rho_{\varepsilon u} \sigma_u & \rho_{vu} \sigma_v \sigma_u & \sigma_u^2 \end{pmatrix} \right). \quad (17)$$

This lets the authors recover:

- $\rho_{\varepsilon v}$: the strength of adverse selection;
- the implied distribution of default thresholds,

$$\tilde{C}_{it} | x_i \sim N(-x'_i \beta, 1/\alpha^2). \quad (18)$$

Interpretation. The joint model converts the reduced-form decomposition into objects that can be used in policy simulations.

3.7 Stylized structural model for LTV-cap policy

To study regulation, the paper builds a two-period model in which borrowers choose among contracts $\{L_k, B(L_k)\}$.

The observed part of utility is

$$U_i(L_k) = u(y_0 - (H_0 - L_k)) \quad (19)$$

$$+ \beta \left[\int_h^{B(L_k)-C_i} u(y_1 - C_i) dF(H_1) + \int_{B(L_k)-C_i}^{\bar{h}} u(y_1 + H_1 - B(L_k)) dF(H_1) \right]. \quad (20)$$

A type-I extreme-value preference shock ϵ_i^L implies logit choice probabilities:

$$P_{ik} = \frac{e^{\gamma U_i(L_k)}}{\sum_{k'} e^{\gamma U_i(L_{k'})}}. \quad (21)$$

Expected lender profit on a standard borrower is

$$\pi^s(L_k, B(L_k); C_i) = -L_k + \frac{1}{1 + r_f} \left[\int_h^{B(L_k)-C_i} \delta H_1 dF(H_1) + \int_{B(L_k)-C_i}^{\bar{h}} B(L_k) dF(H_1) \right]. \quad (22)$$

The policy counterfactual compares:

- an **LTV cap of 100**;
- an **LTV cap of 90 with no supply response**;
- an **LTV cap of 90 with full lender supply response**.

Interpretation. This is where the paper turns the empirical adverse-selection estimate into a market-design result: leverage caps change screening and therefore move the whole contract menu.

4 Identification and instruments (what makes the estimates credible)

4.1 Core identification logic

The design uses the fact that option ARM balances depend on the interaction of:

- the **index** written into the contract;
- the **origination month**, which pins down the subsequent path of index rates;
- the product's **negative amortization mechanics**, which pass interest-rate changes into balances.

Because monthly payments are largely fixed in the short run, the resulting variation mainly shifts **current leverage** rather than the immediate payment burden.

What identifies α . Compare borrowers with the same initial leverage choice but different realized leverage because they were exposed to different index paths.

What identifies γ . After isolating the causal effect of realized leverage, the residual role of initial leverage is interpreted as adverse selection.

4.2 Why the exclusion restriction is plausible

The paper’s key assumption is

$$\text{Corr}(Z_{it}, e_{it}) = 0, \tag{23}$$

conditional on controls.

The argument has three parts:

- **Index fixed effects** absorb any fixed differences between LIBOR, Treasury, COFI, and fixed-index borrowers.
- **Origination-month fixed effects** absorb aggregate time-series conditions.
- **Originator $\times$ origination-month fixed effects** absorb time-varying underwriting and borrower sorting at the lender level.

Interpretation. The remaining variation comes from the **interaction** of index and cohort, not from broad differences across indexes or across time.

4.3 Evidence on first stage and balance

The first stage is strong. In the most saturated specification, the coefficient on the leave-one-out mean is about **0.457** with a **Kleibergen-Paap F-statistic of 25.0**. Less saturated specifications produce F-statistics around **26–28**.

The instrument also appears balanced with respect to observables. Conditioning on the minimal fixed effects, it is strongly related to:

- **current leverage** (first stage);
- **default** (reduced form),

while showing no meaningful correlation with:

- FICO score,
- origination CLTV,
- origination home value,
- documentation status,
- investor status,
- purchase versus refinance,
- single-family status.

Interpretation. This does not prove exogeneity, but it is the exact pattern one wants to see if the instrument is moving balances rather than reflecting borrower composition.

4.4 Threats the paper addresses

Payment channel. A possible concern is that initial leverage correlates with minimum payments, so the origination-leverage coefficient might partly reflect payment stress rather than selection. In practice, controlling for minimum payments barely changes the results.

Geographic confounds. The authors show robustness to dropping zip-code fixed effects and to replacing them with much stronger **MSA $\times$ origination-month fixed effects**. Results remain similar.

Prepayment and attrition. The main analysis uses loans active at 24 months. The paper therefore studies full-sample intent-to-treat style specifications using inferred balances, and separately examines prepayment directly. The core decomposition remains intact.

Second liens. Since current leverage is only observed for the option ARM first lien, the authors re-estimate the model on loans with **no junior lien**. Results are similar.

Index composition. Restricting the sample to the two dominant indexes, **LIBOR and Treasury**, also leaves the main message intact.

4.5 What the estimates do and do not mean

What they do mean.

- The coefficient on current leverage is a **causal effect for compliers** affected by the instrument.
- The coefficient on initial leverage, once current leverage is instrumented for, reveals economically meaningful **adverse selection**.

What they do not mean.

- They are not necessarily representative of standard conforming fixed-rate mortgages.
- They do not rule out all remaining unobserved confounds.
- The policy simulations are stylized and do not include broader general-equilibrium effects, such as house-price feedback.

4.6 External-validity caveat

The paper repeatedly emphasizes that option ARM borrowers differ from the broader fixed-rate mortgage population. The sample contains a very large share of **no/low-documentation loans**, and the adverse-selection effect appears especially strong in that segment.

Interpretation. The paper is best read as strong evidence that both moral hazard and adverse selection mattered in a crucial crisis-era mortgage segment, not as a universal parameter estimate for all household credit.

5 Tables and figures

5.1 Figure 1: comparison of mortgage indexes

Read:

- Panel A shows that one-year LIBOR and one-year Treasury rates moved closely before the crisis but diverged sharply in 2007–2008.
- Panel B shows the spread between the two jumping dramatically, peaking above 3 percentage points.
- For otherwise similar option ARMs, that divergence generated large differences in realized balances.

Economic message. This is the paper's natural experiment in one picture. The crisis did not just change rates on average; it changed them differently across indexes, which translated directly into ex post leverage differences.

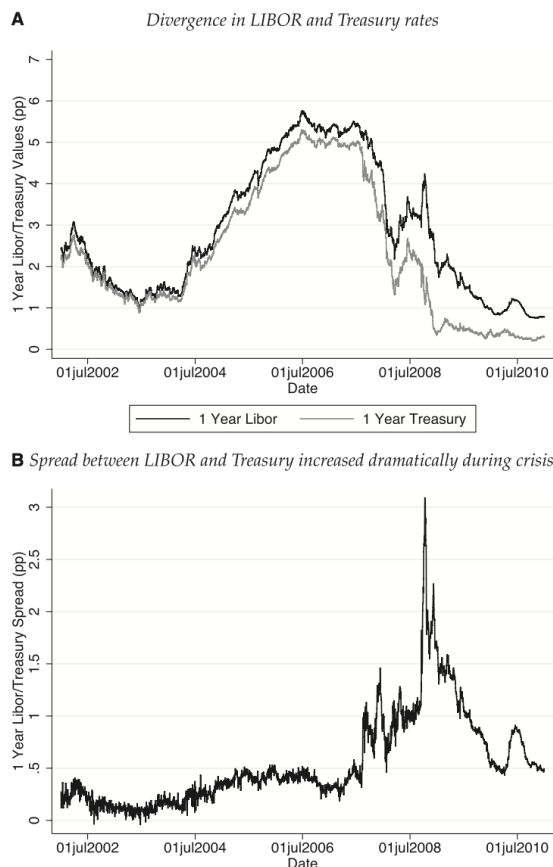


Figure 1
Comparison of mortgage indexes
Panel A shows one-year LIBOR and one-year constant maturity Treasury (CMT) rates between 2002 and 2010.
Panel B plots the spread between the two rates.

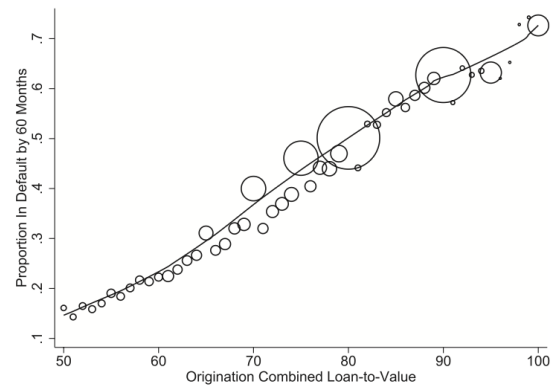


Figure 2
Original CLTV is positively correlated with default within 60 months
Hollow dots show the average proportion of loans defaulting within 60 months for each one-point bin of origination combined loan-to-value. Size of dots is proportional to number of borrowers within each bin. Default is defined as 60 or more days past due. The solid line shows a local linear smoothing of the raw data.

5.2 Figure 2: origination CLTV is strongly correlated with default

Read:

- The raw data show a steep, approximately linear upward relationship between origination combined LTV and default within 60 months.
- Defaults are modest for borrowers near 50–60 CLTV but exceed 50% for the highest-leverage borrowers.
- This is the basic correlation the paper wants to decompose.

Economic message. The unconditional leverage-default relationship is large and obvious. The paper's contribution is not to document this correlation, but to separate it into **selection** and **causal leverage effects**.

5.3 Figure 3 and Table 2: the instrument is uncorrelated with observables

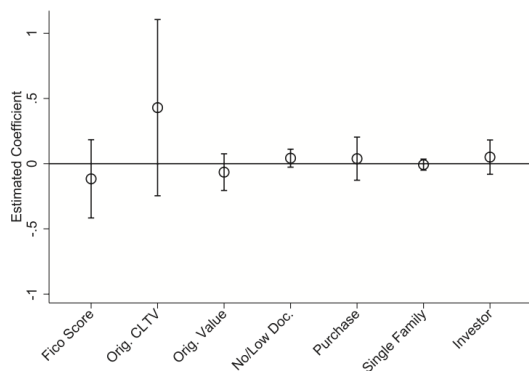


Figure 3

Instrument uncorrelated with observables at origination

This figure shows coefficients from a regression of our leave-out-mean instrument Z_{it} on a range of borrower-level and loan-level covariates. We control throughout for our minimal set of controls: origination month, index type, and zip code. We cluster standard errors at the originator level. The first three variables—FICO score, origination LTV, and origination home value—are also normalized by dividing by their standard deviations to provide interpretable coefficients. The last four variables are dummy variables indicating the presence of no or low documentation in loan origination, whether the mortgage is for a home purchase instead of a refinancing, whether the property is single family, and whether the borrower was a reported investor. Across all categories, we find no statistically significant relationship between our instrument and borrower characteristics. Regression results are also displayed in Table 2.

Table 2 Instrument uncorrelated with observables at origination		Orig. FICO	Orig. CLTV	Orig. value	No/low doc.	Purchase	Single family	Investor
Dep. var.	Current LTV	Default						
Current LTV Leave-Out Mean (Index $\times$ Orig. Month)	0.843*** (0.183)	0.536*** (0.183)	-27.667 (36.435)	0.224 (0.180)	-0.988 (1.090)	0.225 (0.187)	0.206 (0.450)	-0.038 (0.116)
Orig. Month FEs	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Index Type FEs	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Zip code FEs	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Mean of Dep. Var.	0.99	0.44	706.5	0.81	4.48	0.84	0.33	0.64
N	491,215	491,215	433,809	491,215	491,215	491,215	491,215	491,215

This table shows coefficients from a regression of our leave-out-mean instrument Z_{it} on a range of borrower-level and loan-level covariates and outcomes. We include all loans in our sample that are active at 60 months. We control throughout for our minimal set of controls: origination month, index type, and zip code. The first two columns represent basic first-stage and reduced-form versions of our IV specification, respectively. In the first, current leverage, measured as current LTV on the option ARM, is included as a dependent variable. In the second, default, defined as falling 60 days past due between 24 and 36 months, is included as a dependent variable. FICO score, origination LTV, and origination home value are all continuous, while the last four variables are dummy variables indicating the presence of no or low documentation in loan origination, whether the mortgage is for a home purchase instead of a refinancing, whether the property is single family, and whether the borrower was a reported investor. Mean of Dep. Var. refers to the mean of the dependent variable. We cluster standard errors at the originator level. * $p < .1$, ** $p < .05$, *** $p < .01$.

Read:

- The coefficients on borrower observables are all close to zero once index, origination month, and zip-code fixed effects are included.
- Table 2 shows that the same instrument strongly predicts **current LTV** and **default**, but not FICO, initial leverage, house value, documentation, occupancy, or investor status.
- This is exactly the balance pattern needed for the IV design to be believable.

Economic message. The leave-one-out instrument appears to move the **leverage state** of the borrower rather than the borrower's ex ante observable risk profile.

5.4 Table 3: main linear decomposition of default

Read:

- In the baseline specification, the coefficient on **origination leverage** is about **1.066**. A 10-point increase in initial leverage is associated with roughly a 10.7 percentage-point increase in one-year default risk.
- In the richest OLS specification, the coefficient on origination leverage falls to about **0.818**, while the coefficient on **current leverage** is about **0.285**.
- In the richest IV specification, the coefficient on origination leverage is about **0.655**, while the coefficient on current leverage rises to about **0.473**.
- The first stage remains strong, with a coefficient of about **0.457** and an F-statistic of **25.0**.

Economic message. Instrumenting for current leverage makes the causal effect larger and still leaves a large role for initial leverage. That is the paper’s central evidence that both **moral hazard** and **adverse selection** are present, with selection explaining roughly **60%** of the correlation.

Table 3

Impact of origination and current leverage on default: linear model

A. Estimates of origination and current leverage on default (60 DPDs)

	Baseline			OLS			IV		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Origination Leverage	0.917*** (0.064)	1.062*** (0.067)	1.066*** (0.073)	0.666*** (0.064)	0.822*** (0.061)	0.818*** (0.075)	0.472** (0.182)	0.761*** (0.198)	0.655*** (0.127)
Current Leverage				0.301*** (0.048)	0.278*** (0.058)	0.285*** (0.058)	0.533** (0.225)	0.427* (0.247)	0.473*** (0.171)
Orig. Month FEs	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Index Type FEs	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Zip code FEs	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Credit/Loan Controls	No	Yes	Yes	No	Yes	Yes	No	Yes	Yes
Originator × Orig. Month FEs	No	No	Yes	No	No	Yes	No	No	Yes
Mean of Dep. Var.	0.44	0.44	0.44	0.44	0.44	0.44	0.44	0.44	0.44
N	491,215	491,215	491,215	491,215	491,215	491,215	491,215	491,215	491,215

B. First-stage estimates of current leverage on leave-out mean

Current Leverage Leave-Out Mean (Index × Orig. Month)	0.658*** (0.129)	0.648*** (0.122)	0.457*** (0.091)
Kleibergen-Paap F-Statistic	26.1	28.1	25.0

Panel A shows our main results, regressions of default between 24 and 36 months on origination and current leverage. We include a cross-section of all option ARM's in our sample active at 24 months with an origination combined loan-to-value between 50 and 100. Default is defined as falling 60 days past due between 24 and 36 months. Origination leverage is defined as the combined loan-to-value ratio at origination. Current leverage is defined as the current loan-to-value ratio on the option ARM at 24 months. All specifications control for origination month, index type, and zip code fixed effects. Credit/loan controls refer to the original home value (we include both a linear term and dummies for each \$25k bin), dummies for each 20-point bin of FICO credit score at origination, and categorical dummies for occupancy, property type, loan purpose, and documentation level. Originator × orig. month fixed effects refer to fixed effects for each combination of originator and origination month. In columns labeled IV we instrument for current leverage with Z_{it} , the leave-out mean of current leverage at 24 months for all loans with the same origination month and index type. Panel B shows coefficients and F-statistics from first-stage regressions. Mean of Dep. Var. refers to the mean of the dependent variable. * $p < .1$; ** $p < .05$; *** $p < .01$.

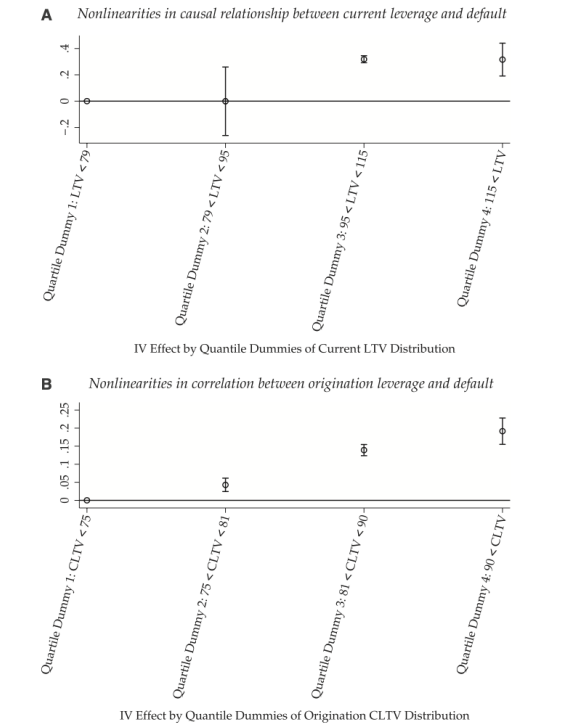


Figure 5
Nonlinearities in the relationship between leverage and default
 Both panels plot coefficients from the regressions displayed in column (9) of Appendix Table A.5, which are described in Equation 6. Panel A shows the coefficients on dummy variables of each quartile of current LTV, as instrumented by quartiles of our leave-out mean Z_{it} . Each coefficient can be interpreted relative to the omitted dummy variable—borrowers with a current $LTV < 79$. Panel B shows coefficients on dummy variables corresponding to quartiles of origination leverage, measured as the origination combined LTV. The effect for these specifications, as well, is normalized so that the first quartile has an effect of 0. The sample includes all active loans at 24 months. Default is defined as falling 60 days past due between 24 and 36 months. Regressions control for index type, zip code fixed effects, originator $\times$ origination month fixed effects, and our full set of credit and loan controls. These controls include the original home value (we include both a linear term and dummies for each \$25k bin), dummies for each 20-point bin of FICO credit score at origination, and categorical dummies for occupancy, property type, loan purpose, and documentation level. Standard errors are clustered at the originator level.

5.5 Figure 5: nonlinear causal effects of leverage

Read:

- Panel A shows almost no causal effect when current LTV is below about 95, then a large jump

once borrowers move into the 95–115 range.

- The estimated effect for the 95–115 quartile is about **31.8 percentage points** relative to the lowest quartile, while the effect for the >115 quartile is nearly the same (about **31.6 points**).
- Panel B shows a much more monotonic, close-to-linear relationship between **origination leverage** and default, consistent with adverse selection across the initial contract choice.

Economic message. The causal effect of leverage is concentrated around the **negative-equity threshold**. Once borrowers are already deep underwater, even more leverage adds relatively little. This supports a **double-trigger** view in which negative equity is necessary but not sufficient for default.

5.6 Table 6 and Figure 6: policy simulations with an LTV cap

Table 6
Simulation results: the impact of a reduction in the LTV cap from 100 to 90

	Col. 1: LTV cap of 100	Col. 2: LTV cap of 90 No supply response	Col. 3: LTV cap of 90 With supply response
Average Loan Size (Thousands)	\$387.5	\$361.4	\$361.3
Average Interest Rate	14.82%	11.38%	11.83%
Average Balance (Thousands)	\$445.0	\$402.5	\$404.0
Defaults	18.31%	15.09%	15.19%
Naive Defaults	-	14.24%	-
Parameters			
Initial Price: $H_0 = \$495.0$		Final Value: $H \sim N(\mu_1 = 594, \sigma_1 = 297)$	Proportion Behavioral: 1%
CARA Coefficient: $a = 0.005$		Viscosity: $\gamma = 1.075$	$\beta = \frac{1}{1 + \gamma f} = 0.975$
Prop. Recovered in Default: $\delta = 0.80$		$C_i \sim N(0.38\mu_1, 0.43\mu_1)$	Borrowers: $N = 1000$

Simulations from structural model described in Section 4. CARA utility assumed, 1,000 simulated borrowers, with 1% behavioral. Viscosity set to match estimated $\rho = 0.118$. Parameters of C_i distribution taken from estimation in Table 5. The mean is based on the linear projection using average value of continuous covariates and the omitted category of all categorical covariates. Initial home prices are set based on average for the largest index category (Treasury). 20% expected net growth over the term of the loan assumed. The first column shows equilibrium outcomes with an LTV cap of 100. The second column shows borrower responses to the removal of all contracts with initial LTV between 90 and 100, holding fixed all contracts with initial LTV less than 90. Naive defaults refers to expected defaults calculated ignoring borrower heterogeneity and extrapolating from default probabilities at each LTV with an LTV cap of 100. The third column shows equilibrium outcomes with an LTV cap of 90.

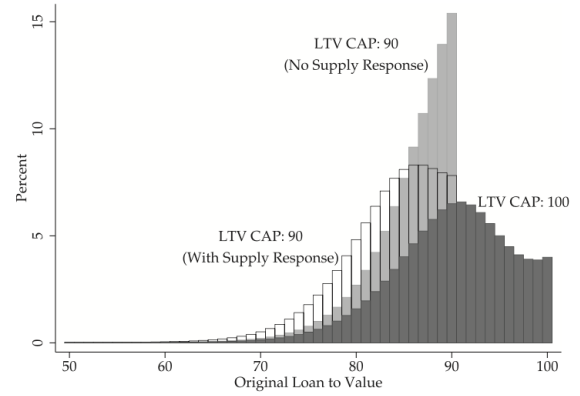


Figure 6
Effect of LTV cap of 90 on leverage: with and without supply response
Bars show simulated proportion of borrowers choosing each original LTV under three regimes. The dark gray bars show equilibrium LTV choices at an LTV cap of 100, the light gray bars show borrowers' LTV choices after a reduction in the LTV cap to 90, but allowing no changes in the prices of contracts below 90. White bars show equilibrium LTV choices with an LTV cap of 90, allowing for the supply response. Figure is based on an exaggerated level of adverse selection. Table 6 shows appropriately calibrated results.

Read:

- With an LTV cap of 100, average loan size is about **\$387.5k**, the average interest rate is **14.82%**, and defaults are **18.31%**.
- If the cap falls to 90 *without* a supply response, average loan size falls to about **\$361.4k** and defaults fall to **15.09%**; a naive regulator ignoring selection would predict an even lower default rate of **14.24%**.
- With a full supply response, the average interest rate rises from **11.38%** to **11.83%** relative to the no-supply-response case, and the final balance rises from about **\$402.5k** to **\$404.0k**.

Economic message. Ignoring adverse selection makes LTV caps look too effective and too cheap. In equilibrium, screening distortions push up rates and partially offset the intended default reduction.

Read:

- The dark bars show the original equilibrium with an LTV cap of 100.

- The light bars show the naive “no supply response” benchmark after a cap of 90.
- The white bars show the true equilibrium with supply response: borrowers shift farther left because risky types compressed into the 90-LTV region raise prices there, pushing even safer borrowers toward smaller loans.

Economic message. Leverage is a **sorting device**. When regulation shuts down the top of the leverage distribution, the entire separating structure of the market moves.

6 Short summary

- The paper uses option ARMs to separate **current realized leverage** from the **initial leverage choice**.
- A crisis-era divergence between LIBOR and Treasury rates creates plausibly exogenous variation in borrower balances.
- IV estimates imply that a 10-point increase in current LTV raises default by roughly **4.7 percentage points**, showing an important causal role for leverage.
- Even after controlling for realized leverage, initial leverage still strongly predicts default, implying economically large **adverse selection**.
- The best benchmark decomposition is roughly **60% adverse selection** and **40% moral hazard**.
- The causal effect is highly nonlinear and concentrated near the threshold of negative equity.
- A stylized model shows that LTV caps reduce defaults less than a naive regulator would expect because leverage also screens borrower risk in equilibrium.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

The paper establishes that the leverage-default relationship in this crisis-era mortgage market reflects **both** a causal role for leverage and substantial adverse selection. The empirical contribution is to separately identify those two forces using option ARM contract mechanics and the unexpected divergence of mortgage indexes during the financial crisis.

7.2 Economic intuition

A borrower can default more because leverage is high today, but borrowers who most value leverage also appear to be riskier ex ante. In equilibrium, leverage therefore has a double role: it shapes **ex post incentives** and also **screens types**. That is why leverage regulation affects the full menu of mortgage contracts, not just the riskiest borrowers.

7.3 Takeaway

Selection, Leverage, and Default in the Mortgage Market gives a clean empirical decomposition of one of the central facts of the housing crisis. The main lesson is that mortgage leverage should not be treated as a purely mechanical driver of default. It is also part of an asymmetric-information problem, and policy must account for both sides at once.